

Response to the Editor and Reviewers

Submission ID: 20e66410-c472-42dc-ab99-957c24a43f96

Original title: Auditing observability bias in InSAR-derived subsidence exposure estimates with a reproducible geospatial workflow

Revised title: Spatial aggregation alters population-weighted estimates of InSAR observation support

We thank the Editor and Reviewers for their comments. We have clarified the research question, added comparisons of allocation and observation support, and withdrawn the interpretation of the original 3.77-million figure as verified subsidence exposure. Below we describe the changes and supporting results. Page references refer to the revised manuscript and Supplementary Information.

Editor

Comment. In particular, the scientific gap and novelty of the study should be more clearly established and supported by a comprehensive and balanced review of the relevant literature. The rationale for the proposed workflow and its methodological choices should be strengthened and better positioned in relation to existing approaches. Further clarification and justification are also required regarding the interpretation and validation of the “hidden exposure” concept, the treatment of observability, the limitations of the GHSL data, and the selection of the InSAR datasets.

Response. We have revised the Introduction to recognize established InSAR quality control and published exposure methods, including recent nationwide, building-level and coastal assessments. The question is now how redistributing a fixed population total within reporting units changes a support summary when the observations and geographic domain remain unchanged. We also explain why each allocation method and dataset was selected.

The added comparisons test aggregation, controlled missingness, population-model dependence and transfer across products and regions. They support an allocation result, rather than a count of residents exposed to subsidence in unobserved locations. The replies below explain the evidence and remaining validation limits.

Location. Main Sections 1–2 (pp. 2–9), Table 2 and Discussion; Supplementary S6 (p. 7) and S31 (p. 32).

Code and supporting files

The scientific reproduction package and experiment guide are available at v0.6.3. It includes processed inputs, reference tables and checksums for the additional analyses; provider-controlled inputs and raw processing stacks have documented acquisition routes. `Local_Validation_Audit.zip` supplies the Bangkok feasibility replay, and `SBKK_Support_Audit.zip` reproduces product-support statistics. Neither establishes independent Bangkok accuracy. The current response and submission bundle are archived at v0.6.4; the scientific results are unchanged. The earlier DOI 10.5281/zenodo.21444768 identifies v0.3.0 only; a revised DOI deposit remains pending.

Reviewer 1

R1.1

Comment. The manuscript presents a valuable methodological contribution to the Earth science community by addressing an important but usually overlooked issue »> how InSAR product support masks affect downstream exposure estimates. There are some critical point that must be addressed. 1- The "Hidden Exposure" claim is not validated. The authors should justify such a claim in a scientific way as they address other issues. The manuscript repeatedly emphasizes that it is not validating deformation in hidden cells, yet the central result (3.77 million people in "hidden exposure") is presented as if it carries decision-relevant weight.

Response. We agree that the original 3.772492-million statistic did not validate subsidence exposure. We have withdrawn that interpretation and the associated measurement-priority claim. The title, definitions, Results and Conclusions now concern population-weighted observation support; S2 retains the original calculation for traceability.

The revised comparison holds the population model, domain and support field fixed. Native GHSL allocation gives a mean area-pair support deficit of 3.020 million allocation units, compared with 4.763–4.770 million under uniform 1-km allocation. These averages are not counts of particular residents. Geometry, conservation and controlled masking check the calculation; excluding boundary cells retains a 58.24–58.56% difference.

For the actual binary final mask, S13 gives a threshold-class share of [0.2797, 0.5741]% at LOS rate $\leq -5 \text{ mm yr}^{-1}$, conditional on observed rates and the population model. This bounds missing class allocation without inferring missing motion; it does not propagate errors in those inputs or use the continuous deficit as a missing-population count.

Independent Bangkok accuracy remains unestablished. The screened GNSS stations share an InSAR pixel, and no independent common datum links their global-frame series to the relative inversion. S17 reports this feasibility result and the separate internal reconstruction checks; S30 reports mixed GNSS agreement for the California products. We distinguish these checks from physical validation of the withdrawn exposure claim.

Location. Main Sections 2.1 (p. 5), 3.1 (p. 10), 3.5–3.7 (pp. 14–16) and 4.5 (p. 20); Supplementary S2 (p. 3), S13 (p. 12), S17 (p. 17) and S30 (p. 29).

R1.2

Comment. P 10: The classification of cells as "majority observable" vs. "not-majority-observable" simplifies a continuous problem. I think binary threshold is unsatisfying the goals and scopes you defined before.

Response. We have replaced the majority split in the primary analysis with continuous population-, built-surface- and area-weighted support distributions over [0,1]. Figures 3–4 show the distributions and sampling sensitivity. The mean deficit is checked against the integral of the empirical distribution; the original binary classification remains only as a historical comparison in Supplementary Figure S1.

Coherence cutoffs 0.2, 0.3 and 0.4, unwrapped validity alone, and expanded 28/56/112/224-pair samples quantify sensitivity. Mean support does not distinguish persistent missingness from variable pass frequency. The 28-pair diagnostic is also kept separate from the connected inversion experiment. S21 reports continuous coverage alongside internal reconstruction error.

Location. Main Sections 2.1 (p. 5), 2.3 (p. 6) and 3.2–3.3 (pp. 10–12), Figures 3–4; Supplementary S2 (p. 3), S16 (p. 14) and S21 (p. 21).

R1.3

Comment. P9-10: "GHSL layers are used because they provide consistent public support for population and built-up surface across domains." »> GHSL data limitations are understated and the authors should improve this statements.

Response. We have expanded the GHSL limitations and quantified population-model sensitivity. On the common valid domain, the $q=0.3$ deficit is 13.37% for GHSL 2020 and 23.20% for WorldPop 2020. Equalizing population totals still gives 2.989 versus 5.186 million allocation units. All 16 crossed model-pair-selection-origin cases retain a positive aggregation difference, but it is much smaller for WorldPop.

We report each model's domain, excluded population and normalized results, compare epochs at fixed resolution, and state the within-pixel allocation assumptions. GHSL population and built surface share information, so built weights are not independent validation. Neither population product establishes local household locations, and their differences are not calibrated error bars.

Further comparisons retain substantial local disagreement after total recalibration: held-out WAPE is 33.24–33.85% for GHSL and 32.34–32.98% for WorldPop. The joint domain of the residential-eligibility comparison covers only 89.13–89.22% of complete Census population, below the exploratory 90% diagnostic gate. We therefore report these results as model and allocation sensitivity, with shared Census lineage and incomplete coverage explicit.

Location. Main Sections 2.5 (p. 8), 3.2 (p. 10) and 4.5 (p. 20); Supplementary S18 (p. 18), S20 (p. 20), S22–S27 (pp. 22–25).

R1.4

Comment. there are also some minor issues that must be double-checked by the authors. I recommend major revisions with particular attention to clarifying the weak points and the policy-relevant interpretation of the results too.

Response. We have checked quantities, units, dates, sample counts, captions, source attribution and cross-references. Historical estimates are separated from the revised support statistic, and source coverage is distinguished from sampled support and final processing quality. We removed claims of verified hidden subsidence exposure and a validated measurement-priority ranking; the practical recommendation now concerns reporting allocation assumptions, support and sensitivity.

Location. Main Sections 3–5 (pp. 10–22); Supplementary S1–S3 (pp. 2–4) and S28 (p. 27).

Reviewer 2

The manuscript "Auditing observability bias in InSAR-derived subsidence exposure estimates with a reproducible geospatial workflow" by Zixuan Liu and Wei Xiong discusses a very important topic regarding InSAR products and how they might translate into misleading exposure estimates. While the topic is relevant and the issues raised by the manuscript are valid, I have major concerns about the structure of the paper.

R2.1

Comment. 1- The problem formulation gives too little credit to established InSAR practices. It discusses the problem in the Introduction. For example:

"exposure accounting can convert this missingness into a denominator change."

"A processing team may publish a careful product with valid-support masks, while a downstream analyst may treat the remaining raster or point layer as a complete hazard domain."

"but they rarely audit whether non-observed radar support is being treated as absent"

I find the concern relevant and valid. But the literature review does not systematically show that previous InSAR exposure studies have failed to address areas with missing information. The manuscript claims that "When the resulting deformation map is intersected with population, built-up land or infrastructure layers, analysts often remove cells that are not valid in the radar product before exposure is summarized," without providing evidence from studies that follow this practice. In my opinion, this is the main weakness of the manuscript. If the issue is misuse of correctly documented InSAR products by non-specialist users, this should be stated explicitly and supported

by multiple examples of studies where such problems occur, together with references to state-of-the-art studies that have tried to address the issue. In its current form, the manuscript sometimes reads as if the InSAR community has failed to recognize this limitation. However, it is well accepted in the community that InSAR observations are opportunistic by nature. In some cases, InSAR products are accompanied by quantitative information on the validity and quality of the observations. For example, LiCSBAS provides average coherence, the percentage of valid unwrapped pixels, etc. along with velocity and time-series products. The manuscript acknowledges that experienced InSAR users understand and correctly interpret areas where InSAR does not provide information. Therefore, the scientific gap should be demonstrated specifically at the point where InSAR products are used in downstream applications. Without enough evidence from previous studies, it is difficult to determine whether the paper addresses a genuine unresolved research problem.

Response. We agree that the original Introduction did not substantiate a general claim that earlier InSAR studies mishandled missing support. We have removed those assertions and now recognize established PS, SBAS, distributed-scatterer, LiCSBAS and MintPy practice, including quality diagnostics and geodetic comparisons.

The revised review identifies documented allocation choices: Blackwell et al. (2020) assumed uniform population within communities, while Ohenhen et al. (2025) assigned census-block populations using median InSAR rates. These are methodological choices, not evidence of misuse. We also discuss Xiong et al.’s nationwide exposure assessment, Yang et al.’s building-level observation criteria, and Oelsmann et al.’s coastal population coverage, weighting and uncertainty analysis.

Our narrower question is whether a support summary is preserved when fine-scale population–support association is replaced by coarse allocation. Geodetic validation of displacement does not itself answer that allocation question. The contribution is the evidence from matched inputs, controlled masking and regional comparisons, rather than a new covariance identity or interpolation theory. We report model dependence, small regional effects and sign reversals in controlled-masking experiments, so the findings do not imply a universal correction or general deficiencies in existing practice.

Location. Main Introduction (pp. 2–4), Sections 3.2 (p. 10), 3.5–3.6 (pp. 14–15) and Discussion; Supplementary S3 (p. 4), S6 (p. 7), S14–S15 (pp. 12–13), S18 (p. 18) and S29 (p. 28).

R2.2

Comment. 2- The manuscript explains how the proposed workflow works, but it does not adequately explain why this particular method is chosen. State-of-the-art methods and alternative approaches should be discussed and compared with the proposed method. The Materials and Methods section also contains very few references to previous methodological works. As a result, it is difficult to determine how the proposed approach relates to existing methods and why the specific choices made in the workflow are preferable or necessary.

Response. We have explained the method choices in relation to established areal interpolation and dasymetric mapping, with references to Eicher and Brewer and Mennis. Holding the population model, support field, domain and total fixed, while conserving extensive quantities during transfer, isolates the effect of allocation.

Each comparator represents a different assumption. Center sampling can miss subcell structure; uniform-area allocation omits within-cell population–support covariance; native-model allocation retains the model’s fine structure without validating it. Built weights add ancillary assumptions, complete-case overlays describe the observed domain, and imputation or additional sensors require separate observations of missing motion. We do not claim superiority across different targets.

The archived footprint and 28-pair sample preserve the link to the submitted analysis. The displayed $q=0.3$ is the middle evaluated cutoff, not an optimal quality threshold. Expanded sampling and alternative cutoffs test sensitivity. On the matched LiCSBAS grid, the allocation shift is 3.066–3.189 percentage points for mean input support but only 0.108–0.117 for the final mask.

Controlled masks compare error and availability against known conditional references; wholly un-estimated units remain visible. These tests do not recover deformation in genuinely missing pixels.

Location. Main Sections 2.3 (p. 6), 2.5–2.6 (pp. 8–9), Table 2 and Sections 3.5 (p. 14) and 3.7 (p. 16); Supplementary S6–S7 (pp. 7–8), S14 (p. 12), S18–S19 (pp. 18–20) and S31 (p. 32).

R2.3

Comment. 3- The "Iran public InSAR products" is based on relatively old study from 2008. More recent nationwide studies of subsidence in Iran are available, including Haghighi and Motagh, 2024 and Payne et al., 2025. InSAR datasets from both of these studies are publicly available. The manuscript should explain why an older dataset was selected instead of these newer products. I think the use of the 2008 dataset does not invalidate the workflow, but it should be justified.

Response. We have corrected the Iran source attribution and excluded the incomplete branch from quantitative applications. The checksum audit identifies the archived rate and mask as the 2014–2020 product accompanying Haghighi and Motagh (2024), rather than a 2008 nationwide rate product. The unmatched seasonal-amplitude archive is unused.

We acknowledge the availability of the newer products, including Payne et al.'s 2014–2022 regional vertical/east–west velocities and uncertainties. These summary layers have different targets from pair-specific support masks. We no longer claim a completed older-product application or a new/old comparison; the quantitative applications use the declared Chao Phraya and California inputs.

Location. Main Section 4.5, Earlier estimates and source attribution (p. 21); Supplementary S1 (p. 2), S3 (p. 4) and S29–S30 (pp. 28–29).